

Homework 8

Name: Diya Jain

$$1. A = \begin{bmatrix} 1 & -10 \\ 6 & -58 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 6 \\ -10 & -58 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 6 \\ -10 & -58 \end{bmatrix} \begin{bmatrix} 1 & -10 \\ 6 & -58 \end{bmatrix} = \begin{bmatrix} 37 & -358 \\ -358 & 3464 \end{bmatrix}$$

$$\det(A^T A) = 4$$

$$a. \text{Vol}(\mathcal{L}) = \sqrt{\det(A^T A)} = \sqrt{4} = 2.$$

b. To show that $\begin{pmatrix} 0 \\ 2 \end{pmatrix} \in \mathcal{L}$, we need to express it as an integer linear combination of the columns of A

$$x_1 \begin{pmatrix} 1 \\ 6 \end{pmatrix} + x_2 \begin{pmatrix} -10 \\ -58 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$x_1 - 10x_2 = 0 \quad - (1)$$

$$x_1 = 10x_2$$

$$6x_1 - 58x_2 = 2 \quad - (2)$$

$$6(10x_2) - 58x_2 = 2$$

$$60x_2 - 58x_2 = 2$$

$$x_2 = 1, \quad x_1 = 10$$

$$\begin{pmatrix} 0 \\ 2 \end{pmatrix} = 10 \begin{pmatrix} 1 \\ 6 \end{pmatrix} + 1 \begin{pmatrix} -10 \\ -58 \end{pmatrix}$$

which shows that $\begin{pmatrix} 0 \\ 2 \end{pmatrix} \in \mathcal{L}$

$$c. y_1 \begin{pmatrix} 1 \\ 6 \end{pmatrix} + y_2 \begin{pmatrix} -10 \\ -58 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$y_1 - 10y_2 = 1 \quad - (1) \quad y_1 = 10y_2 + 1$$

$$6y_1 - 58y_2 = 0 \quad - (2)$$

$$6(1 + 10y_2) - 58y_2 = 0$$

$$6 + 60y_2 - 58y_2 = 0$$

$$2y_2 = -6$$

$$y_2 = -3$$

$$y_1 = 1 + 10(-3) = 1 - 30 = -29$$

Thus,

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = (-29) \begin{pmatrix} 1 \\ 6 \end{pmatrix} + (-3) \begin{pmatrix} -10 \\ -58 \end{pmatrix}$$

which shows that $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \mathcal{L}$.

d. I have already shown that $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ belong to $\mathcal{L}$, and they correspond to the columns of B .

Therefore, $\mathcal{L}$ can be generated by the columns of B as well.

e. The successive minima $\lambda_1(\mathcal{L})$ and $\lambda_2(\mathcal{L})$ are the lengths of the shortest linearly independent vectors in $\mathcal{L}$.

For the matrix B , the columns $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ are linearly independent vectors in $\mathcal{L}$.

$$\left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\| = 1, \quad \left\| \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\| = 2$$

$$\lambda_1(\mathcal{L}) = 1, \quad \lambda_2(\mathcal{L}) = 2$$

$$2. \quad x_1 = \begin{pmatrix} -4 \\ -5 \\ -2 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}, \quad x_3 = \begin{pmatrix} -4 \\ -3 \\ -1 \end{pmatrix}$$

$$e_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{45}} \begin{pmatrix} -4 \\ -5 \\ -2 \end{pmatrix}$$

$$u_2 = x_2 - \text{proj}_{u_1}(x_2) = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 12/5 \\ 3 \\ 6/5 \end{pmatrix} = \begin{pmatrix} 8/5 \\ 0 \\ -16/5 \end{pmatrix};$$

$$e_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

$$u_3 = x_3 - \text{proj}_{u_1}(x_3) - \text{proj}_{u_2}(x_3) =$$

$$\begin{pmatrix} -4 \\ -3 \\ -1 \end{pmatrix} - \frac{1}{15} \begin{pmatrix} -44 \\ -55 \\ -22 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}$$

$$e_3 = \frac{u_3}{\|u_3\|} = \frac{1}{3} \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}$$